

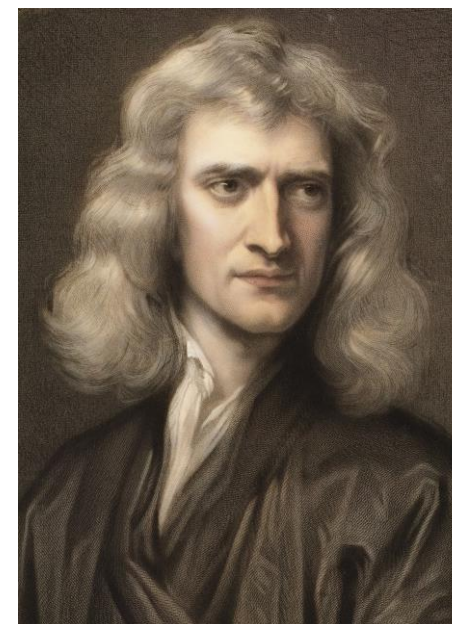
# Math Primer

## (7. Ordinary Differential Equations)

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**Isaac Newton**  
(1643-1727)



**G.W. Leibniz**  
(1646-1716)

# Ordinary Differential Equations

- ▶ Often in physics or engineering, we arrive at a differential equation first and then seek for solutions to the equation. Quant finance is no different. Derivative pricing problems manifest as differential equations (Black Scholes famous PDE)
- ▶ An ordinary differential equation involves ordinary derivatives such as

$$\underline{x}, \quad \underline{y}, \quad \underline{\frac{dy}{dx}}, \quad \underline{\frac{d^2y}{dx^2}}, \quad \dots, \quad \underline{\frac{d^ny}{dx^n}}$$

in an equation  $F(\overset{\uparrow}{x}, y, \underline{y'}, \underline{y''}, \dots, y^{(n)}) = 0$

- ✓ ▶ Order of the DE is the highest order of differentiation in the DE
- ✓ ▶ Degree of the DE is the power of the highest order of differentiation

# Ordinary Differential Equations

$y^4$     $y^{(4)}$

## ► Examples

| DE                                                              | Order | Degree |
|-----------------------------------------------------------------|-------|--------|
| $\left(\frac{dy}{dx}\right)^1 y' = 3y$                          | 1     | 1      |
| $(y')^3 + 4 \ln y = x^3$                                        | 1     | 3      |
| $(y^{(4)})^4 + x^2(y'')^5 + y = 0$                              | 4     | 4      |
| $y'' = \sqrt{y' + y + x}$<br>$\rightarrow (y'')^2 = y' + y + x$ | 2     | 2      |
| $2y'' + x(y')^3 - xy = 0$                                       | 2     | 1      |

# Linear ODE

- ▶ A linear ODE of order  $n$  has
  - ▶ powers of all derivatives = 1
  - ▶ coefficients are only functions of  $x$
  - ▶ only linear term of  $y$  is allowed ( can be thought of as zeroth derivative )

degree = 1

$$\underline{a_n(x)} \left( \frac{d^n y}{dx^n} \right) + \underline{a_{n-1}(x)} \left( \frac{d^{n-1} y}{dx^{n-1}} \right) + \dots + \underline{a_1(x)} \frac{dy}{dx} + \underline{a_0(x)} y = g(x)$$

$y \rightarrow$  '0th derivative of  $y$

# Linear ODE

## ► Examples of linear ODE

| DE                                                                 | Linear ? |
|--------------------------------------------------------------------|----------|
| $\underline{2x}y'' + \underline{x^2}y' - \underline{(x+3)}y = x^2$ | Yes ✓    |
| $\underline{yy''} + xy' + y = 10$                                  | No       |
| $y'' + \sqrt{y'} + y = x^2$                                        | No       |
| $y^{(4)} + \underline{y^4} = 0$                                    | No       |

y

# Solving a DE

- ▶ We can either solve  $y$  in general form with arbitrary constants or solve more explicitly if more conditions are given

$$\text{DE} \rightarrow y = \underline{\underline{f(x)}}$$

- ▶ Exercise : Find the DE by eliminating the constants

$$\left. \begin{array}{l} \text{✓ a) } y = x^3 + \boxed{c}e^{-3x} \rightarrow \\ \text{b) } y = \underline{c_1 e^{-x} + c_2 e^{2x}} \quad \checkmark \end{array} \right\}$$

- ▶ Exercise : Show that  $y$  is a solution to the DE

$$\text{a) } y = \underbrace{(c_1 \sin 2x + c_2 \cos 2x)}_{\uparrow}, \text{ DE : } \underbrace{(y'')}_{\uparrow} + 4y = 0$$

- ▶ Conversely, an  $n$ th order ODE will have a solution with  $n$  arbitrary constants

# IVP vs BVP

- ▶ We can assign specific values to the constants  $(c_1, c_2, \dots, c_n)$  if values of  $(y, y', y'', \dots)$  etc. are given. Of course, one needs  $n$  such conditions to solve for  $n$  arbitrary constants.
- ▶ An **Initial Value Problem** (IVP) is the setup where all values of  $y$  and its derivatives are given for the same  $x$

Example :  $y'' + 2y' = 5e^x$  and  $y(2) = 1$  ,  $y'(2) = 3$

- ▶ A **Boundary Value Problem** (BVP) is the setup where the values of  $y$  and its derivatives are given for different  $x$

Example :  $y'' + 2y' = 5e^x$  and  $y(0) = 1$  ,  $y(1) = 1$

# IVP vs BVP

► Exercise : Which of the following functions is the solution to the IVP

a)  $y_1 = \sin 2x$  b)  $y_2 = x$  c)  $y_3 = \frac{1}{2} \sin 2x$

$\times \uparrow$   $\checkmark$

IVP :  $y'' + 4y = 0$  ;  $y(0) = 0$  ;  $y'(0) = 1$

$\downarrow$   $\downarrow$   $\downarrow$   $\times$

~~$x=0$~~   ~~$x=0$~~

a)  $y_1 = \sin 2x$

$y_1' = 2\cos 2x$

$y_1'' = -4\sin 2x$

$y'' + 4y = 0$

$\checkmark$

b)  $y_2 = x$

$y_2' = 1$

$y_2'' = 0$

c)  $y_3 = \frac{1}{2} \sin 2x$

$y_3' = \cos 2x$

$y_3'' = -2\sin 2x$

$y'' + 4y = 0$

# First order ODE

► First order DE is of the form

$$\underline{y' = f(x, y)}$$

Or could be of the form  $F(x, y, y') = 0$  which can be re-organized as above

Example :

$$\underline{xy' + 2xy - y = 0}$$

$$\Rightarrow \underline{y' = \frac{y - 2xy}{x}}$$

# First order ODE

## ► Type : One Variable Missing

$y$  missing :  $\underline{y'} = \underline{f(x)}$  , solution is  $y = \int f(x)dx$  ✓

$x$  missing :  $\underline{y'} = \underline{f(y)}$  , solution is  $x = \int \frac{1}{f(y)} dy$  ✓

Example :

a)  $\underline{y'} = \underline{y}$  ,  $\underline{y(0) = 1}$  ✓

b)  $\underline{y'} = x^2$  ,  $\underline{y(-1) = 0}$  ↑

a)  $\frac{dy}{dx} = y$   
 $\Rightarrow \int \frac{dy}{y} = \int dx$

$\ln y = x + c$   
 $y = e^{x+c} = e^x \cdot \boxed{e^c} = \overset{k}{\underline{k(e^x)}}$  ,  $k=1$   
 $y = e^x$

b)  $y' = x^2$

$\frac{dy}{dx} = x^2$

$\Rightarrow \int dy = \int x^2 dx$

$y = \frac{x^3}{3} + c$

$0 = \frac{(-1)^3}{3} + c$

$c = \frac{1}{3}$  ,  $y = \frac{x^3 + 1}{3}$

# First order ODE

► Type : **Variable Separable Form**

$$\begin{aligned} y' &= \overbrace{f(x)}^{\downarrow} \overbrace{g(y)}^{\downarrow} \\ \Rightarrow \int \frac{dy}{g(y)} &= \int f(x) dx \end{aligned}$$

Example :

$$y' = \frac{x^2 + 2}{y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2 + 2}{y}$$

$$\Rightarrow \int y dy = \int (x^2 + 2) dx$$

$$\frac{y^2}{2} = \frac{x^3}{3} + 2x + C$$

GS

# First order ODE

► Type : Linear

$$\boxed{\frac{1}{f}} y' + \boxed{P(x)} y = \boxed{Q(x)}$$

$$\int dx = x$$

$$\int dy = y$$

$$\int \underline{d(e^x)} = \underline{e^x}$$

To solve this, multiply Integrating Factor (I.F) =  $e^{\int P dx}$  on both sides. LHS will become an exact differential of  $y \cdot IF$

Example :

a)  $y' - \frac{1}{x} y = x^2$

b)  $\frac{dx}{dy} (1 + ye^x) = e^x$

↗ hw

$$\boxed{\frac{dy}{dx}} \left[ \boxed{-\frac{1}{x}} y \right] = x^2$$

$$P = -\frac{1}{x}, \quad IF = e^{\int -\frac{1}{x} dx}$$

$$e^{\frac{1}{\ln x}} = \frac{1}{x} = e^{-\ln x}$$

$$e^{\ln x} = e^{\ln x^{-1}} = x^{-1} = \frac{1}{x}$$

$$\frac{dy}{dx} - \frac{1}{x} y = x^2$$

$$\frac{1}{x} \frac{dy}{dx} - \frac{1}{x^2} y = x$$

$$\Rightarrow \frac{d}{dx} \left( y \cdot \frac{1}{x} \right) = x$$

$$\int \underline{d\left(\frac{y}{x}\right)} = \int x dx$$

$$\frac{y}{x} = \frac{x^3}{2} + C$$

$$y = \frac{x^3}{2} + Cx$$

# Second order ODE

► Typically, second order ODE (degree 1) takes the form

$$\underline{\underline{y''}} = f(\overset{\uparrow}{x}, \overset{\uparrow}{y}, \overset{\uparrow}{y'})$$

$$\left(\frac{dy}{dx}\right)^1$$

The solution will involve 2 arbitrary constants.

# Second order ODE

► Case I :  $y'$ ,  $y$  are missing

$$\overset{\uparrow}{y''} = \overset{\uparrow}{f(\underline{x})}$$

Integrate  $y$  twice

Example :

$$y'' = 4x$$

↑

$$\frac{d^2 y}{dx^2} = 4x$$

$$\frac{dy}{dx} = \int 4x dx = 2x^2 + C_1$$

$$y = \int (2x^2 + C_1) dx$$

$$\left\langle y = \frac{2x^3}{3} + \underline{C_1 x} + C_2 \right\rangle$$

# Second order ODE

► Case II : y is missing

$$y'' = f(x, y')$$

$$\frac{dt}{dx}$$

Take  $y' = t$ , so the equation becomes  $t' = f(x, t)$

Example :

$$xy'' + 2y' = x^3$$

HW

$$xy'' + 2y' = x^3 \quad y' = t, \quad y'' = \frac{dt}{dx}$$

$$x \frac{dt}{dx} + 2t = x^3$$

$$\frac{dt}{dx} + \left(\frac{2}{x}\right)t = x^2$$

$$P = \frac{2}{x}$$

$$\begin{aligned} I f &= e^{\int P dx} = e^{\int \frac{2}{x} dx} \\ &= e^{2 \ln x} = x^2 \end{aligned}$$

# Second order ODE

► Case III :  $y', x$  are missing

$$\underline{y''} = f(\underline{y})$$

Take  $y' = t$ , and then write  $y''$  =  $\frac{dt}{dx} = \frac{dt}{dy} \frac{dy}{dx} = t \frac{dt}{dy}$ , so the equation becomes  $t \frac{dt}{dy} = f(y)$  which is in variable separable form.

Example :

$$y^3 y'' = 4$$

$$y^3 \cdot \frac{d^2 y}{dx^2} = 4, \quad t = \frac{dy}{dx} \cdot \frac{dt}{dx} = \frac{d^2 y}{dx^2} = \frac{dt}{dy} \frac{dy}{dx} = t \frac{dt}{dy}$$

$$\Rightarrow y^3 \cdot t \frac{dt}{dy} = 4$$

$$\Rightarrow \int t dt = \int \frac{4}{y^3} dy$$

$$\frac{t^2}{2} = -\frac{2}{y^2} + C_1, \quad t = u$$

# Second order ODE

► Case IV :  $x$  is missing

$$y'' = \underline{f(y, y')}$$

Take  $y' = t$ , and then write  $y'' = \frac{dt}{dx} = \frac{dt}{dy} \frac{dy}{dx} = t \frac{dt}{dy}$ , so the equation becomes  $t \frac{dt}{dy} = f(y, t)$  which is a first order ODE.

Example :

$$y^2 y'' = 4y'$$

$$\begin{aligned}
 & t = \frac{dy}{dx} \\
 & y^2 t \frac{dt}{dy} = 4t \\
 & \Rightarrow \int dt = \int 4 \frac{dy}{y^2} \\
 & t = -\frac{4}{y} + c_1 \\
 & \frac{dy}{dx} = -\frac{4}{y} + c_1 = \frac{c_1 y - 4}{y} \\
 & \int \frac{y' dy}{c_1 y - 4} = \int dx = x + c_2 \\
 & \quad \downarrow \frac{1}{c_1} (c_1 y - 4 + 4) \\
 & \quad \frac{1}{c_1} (c_1 y - 4 + 4)
 \end{aligned}$$

# Linear ODEs in general

► General nth order Linear ODE is of the form

$$\underbrace{a_n(x)}_{\text{coefficient}} \underbrace{y^{(n)}}_{\text{derivative}} + \underbrace{a_{n-1}(x)}_{\text{coefficient}} \underbrace{y^{(n-1)}}_{\text{derivative}} + \dots + \underbrace{a_1(x)}_{\text{coefficient}} \underbrace{y^{(1)}}_{\text{derivative}} + \underbrace{a_0(x)}_{\text{coefficient}} \underbrace{y}_{\text{function}} = \underbrace{g(x)}_{\text{function}}$$

Using the operator  $D := \frac{d}{dx}$ ;  $D^2 := \frac{d^2}{dx^2}$ ; ...;  $D^k := \frac{d^k}{dx^k}$  and so  $D^k y := \frac{d^k y}{dx^k}$

$\underline{Dy} = \frac{dy}{dx}$     $D^2 y = \frac{d^2 y}{dx^2}$     $D^k y = \frac{d^k y}{dx^k}$

Now Define Linear Differential Operator

$$\underline{L} = \underline{a_n(x)} \underline{D^n} + \underline{a_{n-1}(x)} \underline{D^{n-1}} + \dots + a_1(x)D + a_0(x)$$

Above DE now can be re-written as  $Ly = g$

$$L(y) = a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots$$

# Linear ODEs

► Note that  $L$  is a linear operator because

$$\begin{aligned} L(y_1 + y_2) &= L(y_1) + L(y_2) \\ L(cy) &= cL(y) \end{aligned}$$

► if  $g(x) = 0$ , the DE is called **Homogeneous**

The general solution to the DE is given by the sum of

- Solution to the homogeneous equation (complementary function  $y_c$ ),  $Ly_c = 0$  and
- Solution to the original DE (particular integral),  $Ly_p = g$

If there are multiple complementary solutions  $(y_1, y_2, \dots, y_n)$ , then linear combination of them  $(\lambda_1 y_1 + \lambda_2 y_2 + \dots + \lambda_n y_n)$  is also a solution

$$\begin{aligned} L &\equiv \frac{d^2}{dx^2} + \frac{d}{dx} \\ L(y_1 + y_2) &= \frac{d^2(y_1 + y_2)}{dx^2} + \frac{d(y_1 + y_2)}{dx} \\ &= \frac{d^2 y_1}{dx^2} + \frac{d y_1}{dx} + \frac{d^2 y_2}{dx^2} + \frac{d y_2}{dx} \\ &= L(y_1) + L(y_2) \end{aligned}$$

$$y = y_p + y_c$$

$$\begin{aligned} Ly &= g \\ Ly_p &= g \\ Ly_c &= 0 \end{aligned}$$

$$L(y_p + y_c) = (Ly_p) + (Ly_c) = g + 0 = g$$

# Linear ODE with constant coefficients

► Consider a homogeneous DE :  $Ly = ay'' + by' + cy = 0$

Try a solution of the form  $y = e^{\lambda x}$ , then

$$y' = \lambda e^{\lambda x} \quad y'' = \lambda^2 e^{\lambda x}$$

$$e^{\lambda x} (a\lambda^2 + b\lambda + c) = 0$$

$$Ly = (aD^2 + bD + c)e^{\lambda x}$$

We get

$$a\lambda^2 + b\lambda + c = 0 \quad (\text{Auxiliary Equation})$$

Case I :  $b^2 - 4ac > 0$ ,  $y = c_1 \exp(\lambda_1 x) + c_2 \exp(\lambda_2 x)$

Case II :  $b^2 - 4ac = 0$ ,  $y = (c_1 + c_2 x) \exp(\lambda x)$   $\lambda_1 = \lambda_2$  (2 fold root)

Case III :  $b^2 - 4ac < 0$ ,  $y = e^{px} (A \cos qx + B \sin qx)$ , where  $p = -\frac{b}{2a}$ ,  $q = \frac{1}{2a}(\sqrt{b^2 - 4ac})$

$$p + iq, p - iq, \quad p = -\frac{b}{2a}, \quad q = \frac{1}{2a}\sqrt{b^2 - 4ac}$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

# Linear ODE with constant coefficients

► Examples :

$$y = e^{\lambda x}; y' = \lambda e^{\lambda x}, y'' = \lambda^2 e^{\lambda x}.$$

a)  $y'' - 3y' - 4y = 0$   $\rightarrow \lambda^2 - 3\lambda - 4 = 0$   
 $\lambda = 4, -1, y = c_1 e^{4x} + c_2 e^{-x}$

b)  $y'' - 8y' + 16y = 0 \rightarrow \lambda^2 - 8\lambda + 16 = 0$   
 $\Rightarrow (\lambda - 4)^2 = 0, 4, 4, y = (c_1 + c_2 x) e^{4x}$

c)  $y'' - 3y' + 4y = 0$

$$\lambda^2 - 3\lambda + 4 = 0$$

$$\lambda = \frac{3 \pm \sqrt{9 - 16}}{2} = \frac{3 \pm \sqrt{7}i}{2}, p = 3/2, q = \frac{\sqrt{7}}{2}$$

$$y = e^{3x/2} \cdot \left( \underset{\uparrow}{A} \cos\left(\frac{\sqrt{7}}{2}x\right) + \underset{\uparrow}{B} \sin\left(\frac{\sqrt{7}}{2}x\right) \right)$$

# General nth order ODE with const. coefficients

► General nth order ODE with constant coefficients

$$\underline{L(y) = 0}$$

$$L = a_n D^n + a_{n-1} D^{n-1} + \dots + a_1 D' + a_0 \underline{D}$$

$a_i \frac{dy}{dx}$  w/o  $y$

Auxiliary Equation becomes

$$\underline{a_n \lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_1 \lambda + a_0 = 0}$$

**Case I** : n distinct real roots ,  $y = \underline{c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} + \dots + c_n e^{\lambda_n x}}$

**Case II** : r repeated real roots ,  $y = \underline{e^{\lambda x} (c_1 + c_2 x + c_2 x^2 + \dots + c_r x^{r-1})}$

**Case III** : r repeated complex roots of the form  $p + iq$ , then

$$y = \underline{e^{px} (\alpha_1 + \alpha_2 x + \alpha_3 x^2 + \dots + \alpha_r x^{r-1}) \cos qx} + \underline{e^{px} (\beta_1 + \beta_2 x + \beta_2 x^2 + \dots + \beta_r x^{r-1}) \sin qx}$$

$(2+3i)^2$

$(2+1)^2 = 0$

$\lambda^2 + 1 = 0 \quad \pm i, \pm i$

(n) -

$\lambda, \lambda, \lambda, \dots$  r times (n-r)

# General nth order ODE with const. coefficients

► Examples :

a)  $y^{(4)} - 5y'' + 6y = 0$

$y = e^{\lambda x}$

$\lambda^2 = s$

$$\frac{\lambda^4 - 5\lambda^2 + 6}{s^2 - 5s + 6} = 0 \quad (\lambda^2 - 2)(\lambda^2 - 3) = 0$$

$$\pm\sqrt{2}, \pm\sqrt{3}$$

b)  $y^{(6)} - 5y^{(4)} = 0$

$$y = c_1 e^{\sqrt{2}x} + c_2 e^{-\sqrt{2}x} + c_3 e^{\sqrt{3}x} + c_4 e^{-\sqrt{3}x}$$

c)  $y^{(4)} + 2y'' + y = 0$

Hw

$$(\lambda - 0)^4 = (\lambda - 0)(\lambda - 0)(\lambda - 0)(\lambda - 0)$$

$$\lambda^6 - 5\lambda^4 = 0$$

$$\lambda^4(\lambda^2 - 5) = 0$$

$$\lambda = \underline{\underline{0, 0, 0, 0}}, \underline{\underline{\sqrt{5}}}, \underline{\underline{-\sqrt{5}}}$$

$$y = (c_1 e^{\sqrt{5}x} + c_2 e^{-\sqrt{5}x} + (c_3 + c_4 x + c_5 x^2 + c_6 x^3)) \cdot e^{0 \cdot x}$$

# Non-homogeneous case – Particular Solution

► Note that the general solution  $y = y_c + y_p$

$\mathcal{L}(y_c + y_p) = \mathcal{L}y_p = g$

$\boxed{\mathcal{L}y = g} \rightarrow y_p$

$\mathcal{L}y = 0 \rightarrow y_c$

The particular-solution (P.I.) can be attempted using Guess Work by assuming the same form as  $g$

$\mathcal{L}y = 0 \times \mathcal{L}y = \textcircled{g}$

- $y_p = ce^{(k)x}$
- Case I** –  $g(x)$  is a polynomial i.e.,  $g(x) = p_0 + p_1x + \dots + p_mx^m$
  - Case II** –  $g(x)$  is exponential i.e.,  $g(x) = Ce^{kx}$  ( provided  $k$  is not a root of A.E. )
  - Case III** –  $g(x)$  is trigonometric e.g.,  $g(x) = \sin ax$  ( provided  $ia$  is not a root of A.E. )
  - Case IV** –  $g(x)$  is a combination of above

# Non-homogeneous case – Particular Solution

► Examples :

$$a) \quad y'' + 3y' + 2y = 3e^{5x}$$

$$b) \quad y'' + 3y' + 2y = x^2$$

$$c) \quad y'' - 5y' - 6y = \cos 3x$$

# Linear ODE with Variable Coefficients – Euler Equation

► Cauchy Euler equation is given as

$$Ly = \underbrace{(ax^2)}_{\text{coeff}} \frac{d^2 y}{dx^2} + \underbrace{bx}_{\text{coeff}} \frac{dy}{dx} + \underbrace{cy}_{\text{coeff}} = g(x)$$

$$\begin{aligned} y &= x^\lambda \\ y' &= \lambda x^{\lambda-1} \\ y'' &= \lambda(\lambda-1)x^{\lambda-2} \\ a \lambda(\lambda-1)x^{\lambda-2} &+ b \lambda x^{\lambda-1} + c x^\lambda = 0 \\ x^\lambda (a \lambda(\lambda-1) + b \lambda + c) &= 0 \end{aligned}$$

Note that the coefficient has the same power of  $x$  as the order of the derivative

The equation is linear. To solve the homogeneous part  $Ly = 0$ , we try  $y = x^\lambda$ . Upon substitution, it yields the **A.E.**  $a\lambda^2 + b\lambda + c = 0$

✓ **Case I** :  $b^2 - 4ac > 0$ ,  $y = Ax^{\lambda_1} + Bx^{\lambda_2}$  ✓

✓ **Case II** :  $b^2 - 4ac = 0$ ,  $y = (A + B \ln x) x^\lambda$

✓ **Case III** :  $b^2 - 4ac < 0$ ,  $y = x^p (A \cos(q \ln x) + B \sin(q \ln x))$ , where  $p = -\frac{b}{2a}$ ,  $q = \frac{1}{2a}(\sqrt{b^2 - 4ac})$

# Linear ODE with Variable Coefficients – Euler Equation

► Examples :

a)  $x^2 y'' - 2xy' - 4y = 0$

b)  $x^2 y'' - 7xy' + 16y = 0$

c)  $x^2 y'' - 3xy' + 13y = 0$

$$y = \underline{x^\lambda}, \quad y' = \lambda \underline{x^{\lambda-1}}, \quad y'' = \lambda(\lambda-1) \underline{x^{\lambda-2}}$$

$$(a) \quad x^\lambda (\lambda(\lambda-1) - 2\lambda - 4) = 0$$

$$\lambda^2 - 3\lambda - 4 = 0$$

$$(\lambda - 4)(\lambda + 1) = 0$$

$$4, -1$$

$$y = Ax^4 + \frac{B}{x}$$

hw

$$(b) \quad x^\lambda (\lambda(\lambda-1) - 7\lambda + 16) = 0$$

$$\lambda^2 - 8\lambda + 16 = 0, \quad \lambda = 4, 4$$

$$y = (A + B \ln x) x^4$$